

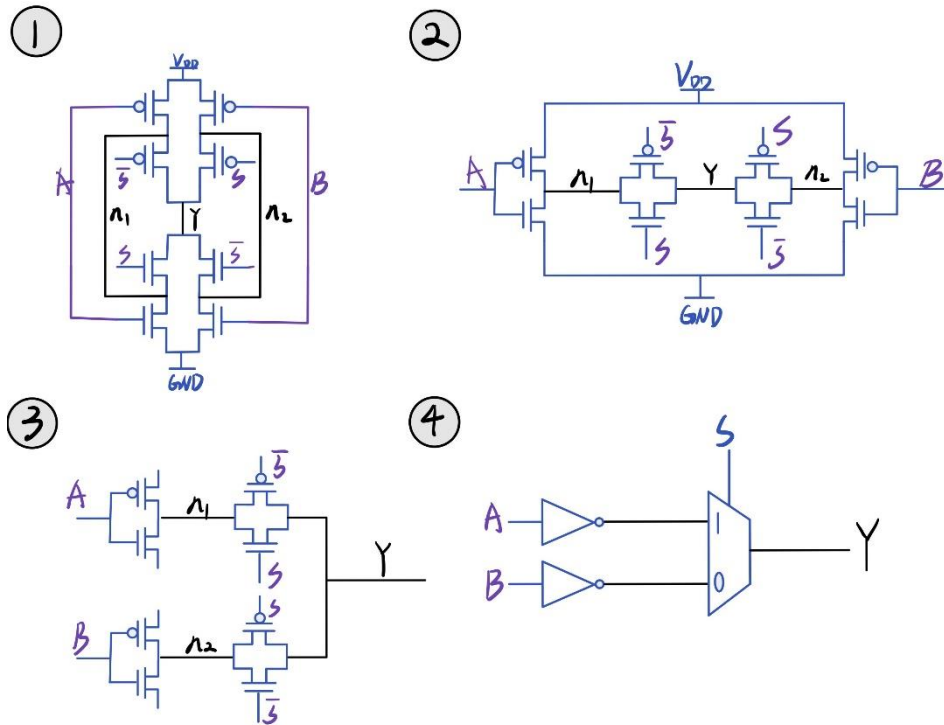
2024 Digital Intergrated Circuits HW2

Given $V_{DD}=0.8V$ · $W_{min}=64nm$ · $L_{min}=32nm$ with resolution of 1nm; Use Low V_t CMOS in this homework. Please submit your SPICE to E3.

(1) Layout (40%)

A. Draw the circuit schematic of the attached layout (a) and (b). The signal name each node shall be marked at the schematic. Explain which circuit has higher speed. (20%)

(a)



上方呈現的圖示乃是版圖(a)所對應的電路示意圖。首先，輸入信號 A 和 B 各自連接至一個反相器(inverter)。這兩個經過反相處理的信號隨後匯流至一個多工器(multiplexer，簡稱 mux)。該多工器由信號 s 操控，決定最終輸出是經反相的 A 信號抑或是經反相的 B 信號。

綜觀整體設計，我們可以清楚地辨識出，(a)所展現的數位邏輯實為一個反相多工器(inverting multiplexer)。

(b)

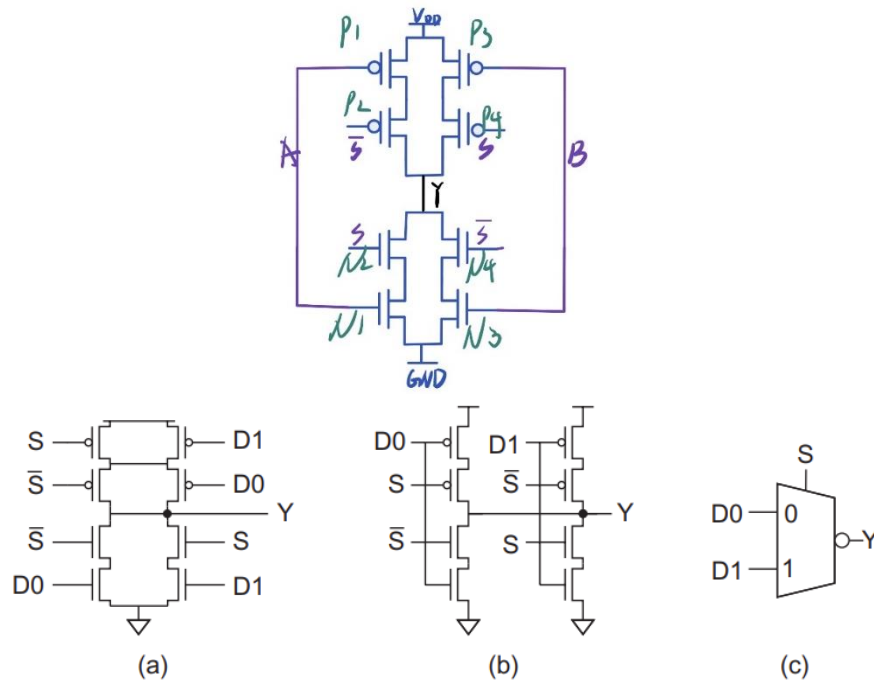


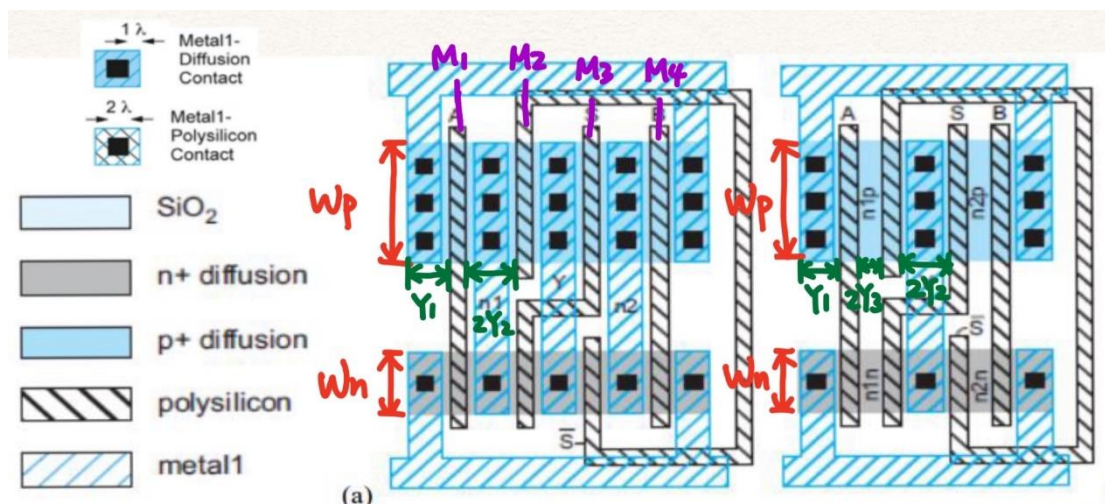
FIGURE 1.29 Inverting multiplexer

Source: Neil H. E. Weste, David Money Harris 講義

Layout(a)與 layout(b)僅差在少了 $n1$ 與 $n2$ 這兩條線，從 circuit schematic 可以看出多了 $n1/n2$ 這條線，從 a/b 訊號輸入時，會需要多推一個 transistor，根據 Electrical Effort 來看的話， h 會增加導致整個 total delay 也會跟著增加，因此 layout(b)擁有更快速的邏輯。

B. Indicate the W/L ratio of each transistors in both (a) and (b) and AD · AS in terms of λ . (10%)

(a)



$$\text{AS of } M_{1P} \& M_{4P} = W_n * Y_1 = 14\lambda * 5\lambda = 70\lambda^2$$

$$\text{AS of } M_{2P} \& M_{3P} = W_p * Y_2 = 14\lambda * \frac{6\lambda}{2} = 42\lambda^2$$

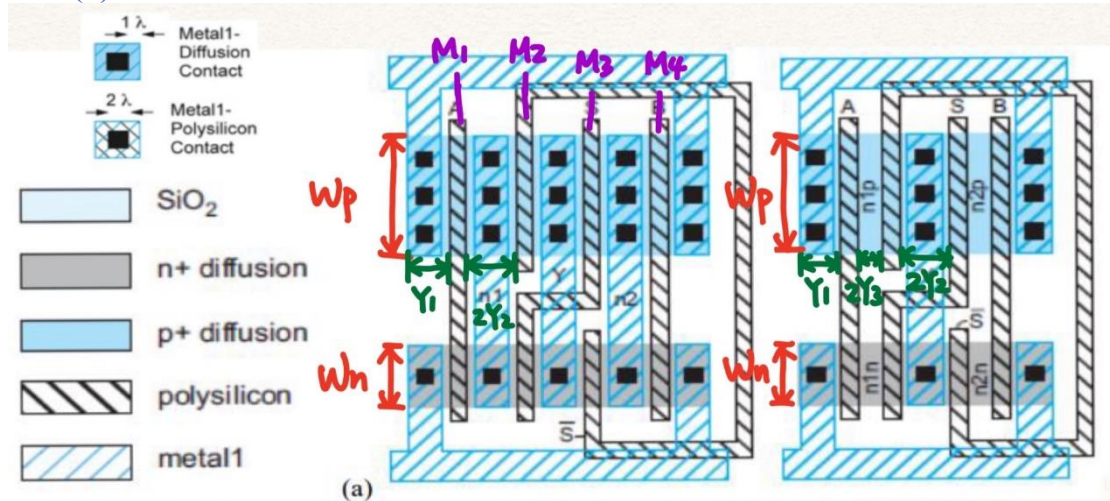
$$\text{AS of } M_{1N} \& M_{4N} = W_n * Y_1 = 4\lambda * 5\lambda = 20\lambda^2$$

$$\text{AS of } M_{2N} \& M_{3N} = W_n * Y_2 = 14\lambda * \frac{6\lambda}{2} = 12\lambda^2$$

$$\text{AD of } M_{1P} \& M_{2P} \& M_{3P} \& M_{4P} = W_p * Y_2 = 14\lambda * \frac{6\lambda}{2} = 42\lambda^2$$

$$\text{AD of } M_{1N} \& M_{2N} \& M_{3N} \& M_{4N} = W_n * Y_2 = 4\lambda * \frac{6\lambda}{2} = 12\lambda^2$$

(b)



$$\text{AS of } M_{1P} \& M_{4P} = W_p * Y_1 = 14\lambda * 5\lambda = 70\lambda^2$$

$$\text{AD of } M_{1P} \& M_{4P} = \text{AS of } M_{1P} \& M_{3P} = W_p * Y_3 = 14\lambda * \frac{3\lambda}{2} = 21\lambda^2$$

$$\text{AD of } M_{2P} \& M_{3P} = W_p * Y_2 = 14\lambda * \frac{6\lambda}{2} = 42\lambda^2$$

$$\text{AS of } M_{1N} \& M_{4N} = W_n * Y_1 = 4\lambda * 5\lambda = 20\lambda^2$$

$$\text{AD of } M_{1N} \& M_{4N} = \text{AS of } M_{2N} \& M_{3N} = W_n * Y_3 = 4\lambda * \frac{3\lambda}{2} = 6\lambda^2$$

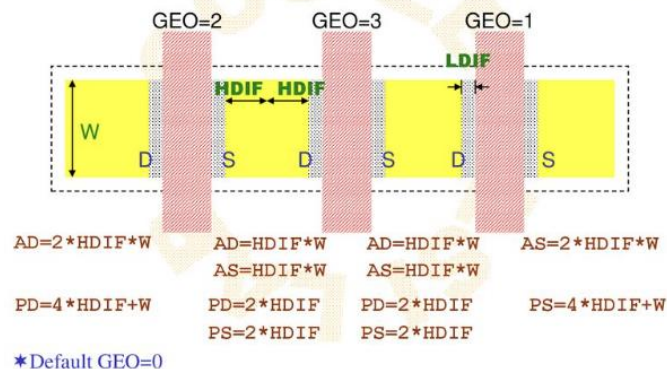
$$\text{AD of } M_{2N} \& M_{3N} = W_n * Y_2 = 4\lambda * \frac{6\lambda}{2} = 12\lambda^2$$

AD, AS 在共用的時候各為面積的一半，

Source: <https://slideplayer.com/slide/15140080/> (slide page.7)

ACM : Area Calculation Method

ACM=3

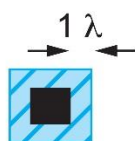


C. Mark the design rules in (a) and (b) that determine the width of this layout. (10%)

在(a)、(b)兩張圖中，都共用以下 Width： $W_p = 14\lambda$ 、 $W_n = 4\lambda$ ，其中 W_p 用以表示 PMOS、 W_n 則用以表示 NMOS。

在本設計中，我們遵循了四項 Design Rule：

1. 打入的 Contact 與金屬層 (Metal) 邊界距離要求：打入的 Contact 的每一邊與其對應的金屬層邊界之間必須保持至少 1λ 的距離。

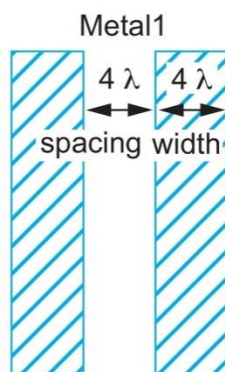


Metal1-
Diffusion
Contact

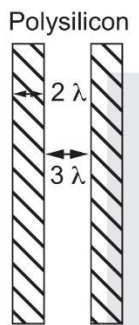
2. Contact 的尺寸規格：每個 Contact 的標準尺寸為 $2\lambda * 2\lambda$ 。此外，相鄰 Contact 之間需要保持至少 3λ 的間距。

● Polysilicon and contacts have a spacing of 3λ from other polysilicon or contacts.

3. 金屬層的寬度和間距規定：金屬層的標準寬度被設定為 4λ 。同時，相鄰金屬層之間必須保持至少 4λ 的間距。

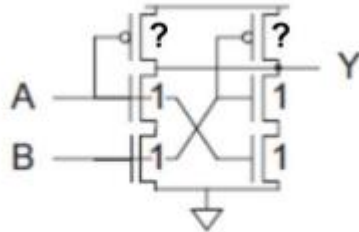


4. 多晶矽 (Polysilicon) 的寬度和間距要求：多晶矽結構的標準寬度為 2λ 。相鄰的多晶矽結構之間需要維持至少 3λ 的間距。



(2) Timing and Power Analysis (60%)

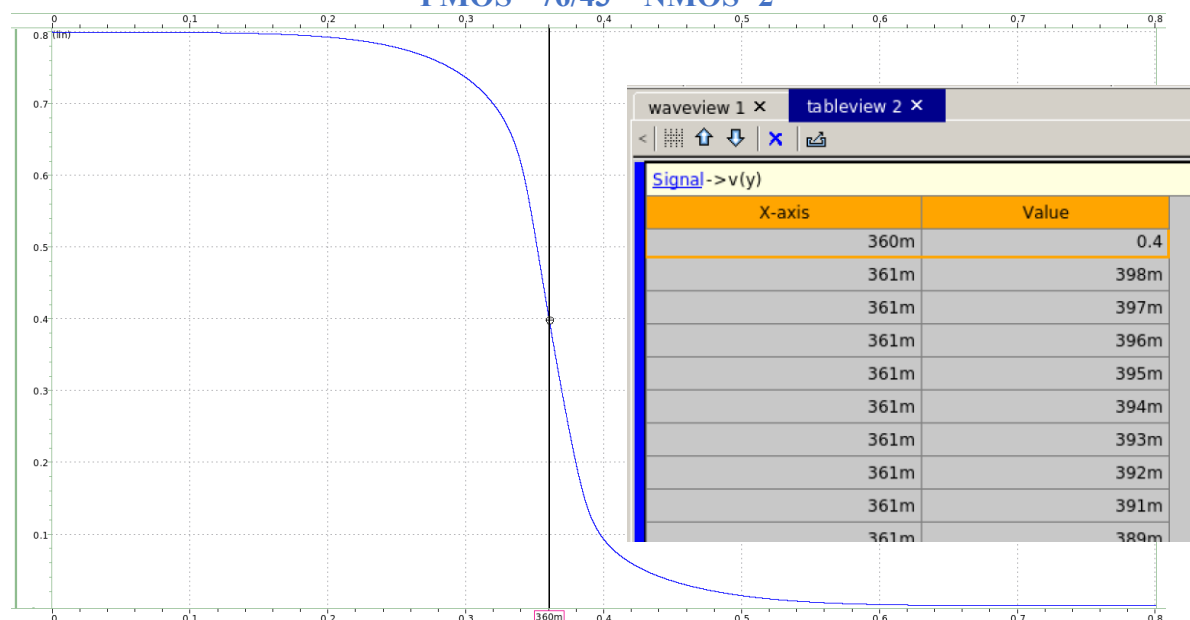
A. Design a two inputs NAND logic gate as shown in the following figure such that it has the logic threshold of $0.45 \cdot V_{DD}$ (when two inputs have the same voltage). Indicate the W/L of PMOS and NMOS and show the SPICE simulation results of VTC. (15%)



```
.param wp = 76n
.param wn = 64n
.param lp = 43n
.param ln = 32n
```

Fig. 2: Perfectly Symmetric 2-input NAND gate

W/L of PMOS and NMOS:
PMOS = 76/43 NMOS = 2



我採取了一種漸進式的方法。初始階段，我將所有元件尺寸設置為最小值，然後開始調整 PMOS 的寬度(wp)。隨著 wp 的增加，電壓轉移特性曲線(VTC)呈現向右偏移的趨勢。這種現象的根源在於:wp 越大，PMOS 在邏輯'1'狀態下的導通能力就越強。

基於這一觀察，為了將 VTC 整體向左調整，我的策略是適度降低 PMOS 的導通能力。為此，我開始調整 PMOS 的長度(lp)。我將 lp 從 32 奈米逐步增加到 43 奈米。在這個過程中，我發現當 lp 落在 40 到 43 奈米之間時，電路性能開始接近題目要求的規格。

最後經過一系列細緻的微調，我終於找到了最佳配置:當 wp 設定為 76 奈米，lp 設定為 43 奈米時，電路的表現完美符合題目的所有規格要求。

B. Run SPICE to get the followings: (25%)**i. output capacitances and input capacitance of the NAND circuit.**

```

nodal capacitance table

node      =   cap      node      =   cap      node      =   cap
+0:a      = 198.2086a  0:b       = 198.2086a  0:vdd     = 376.7394a
+0:vss    = 506.6126a  0:y       = 261.4096a  1:x1      = 160.9710a
+1:x2     = 160.9710a

```

ii. Leakage power dissipation when $V_{in} = \text{logic 1}$ and $V_{out} = \text{logic 0}$.

```

**** voltage sources

subckt
element  0:va      0:vb      0:vvdd      0:vvss
volts    800.0000m  0.      800.0000m  0.
current  -169.9088p -84.9543p -3.3373n   3.5072n
power    135.9270p  0.      2.6699n   0.

total voltage source power dissipation= 2.8058n watts

```

iii. Leakage power dissipation when $V_{in} = \text{logic 0}$ and $V_{out} = \text{logic 1}$.

```

**** voltage sources

subckt
element  0:va      0:vb      0:vvdd      0:vvss
volts    0.      0.      800.0000m  0.
current  39.6307p  19.8154p -13.6646n  13.6250n
power    0.      0.      10.9317n   0.

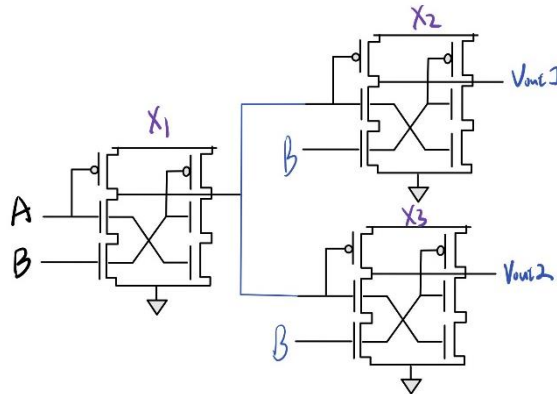
total voltage source power dissipation= 10.9317n watts

```


C. Do the timing analysis of the NAND circuit which has a fanout of two NANDs. Input signal is a pulse waveform with duration of 2ns (with $t_r = t_f = 0.4\text{ns}$ defined as 0%-100% · duty = 0.5) Run two cases by using SPICE (5ns) to get the timing (tdf and tdr) and power waveform: (20%)

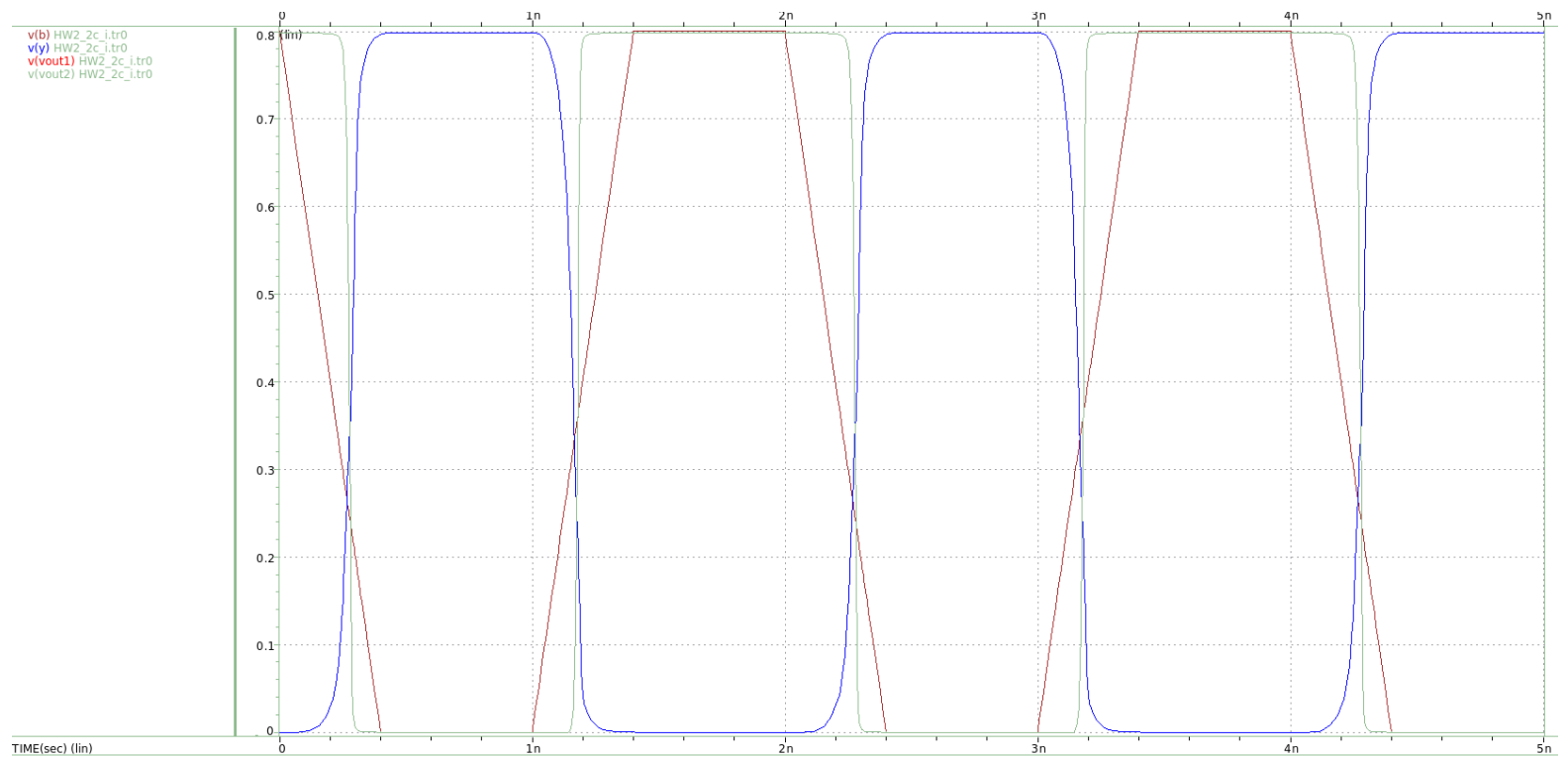
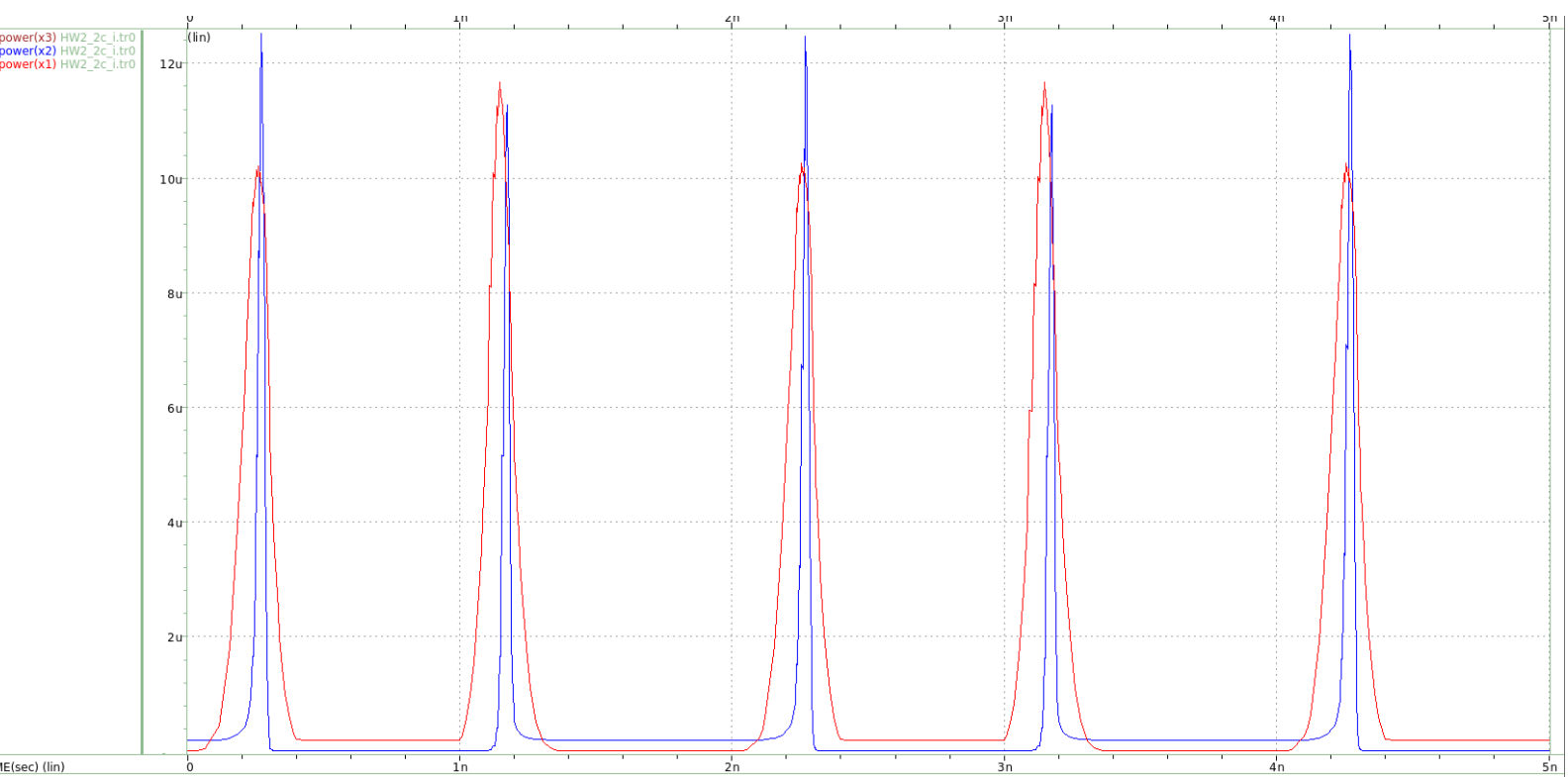
List in table form with the tdf,tdr,average power and peak power of the NAND circuits.

i. A=1 and B changes from 0 to 1.

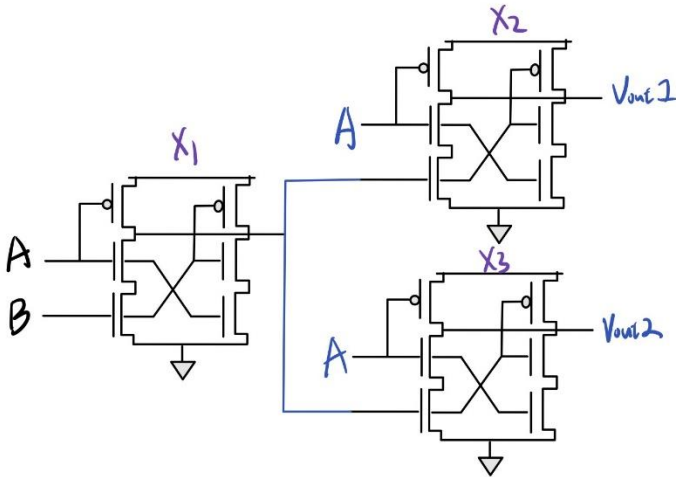


t_df1	t_dr1	t_df2	t_dr2
t_df3	t_dr3	avg_power1	peak_power1
avg_power2	peak_power2	avg_power3	peak_power3
temper	alter#		
-1.924e-11	6.525e-11	-3.988e-12	1.611e-11
-3.988e-12	1.611e-11	1.362e-06	1.167e-05
4.505e-07	1.251e-05	4.505e-07	1.251e-05

Tdf1	-1.924e-11 ns
Tdr1	6.525e-11 ns
Average Power1	1.361e-06 watt
Peak Power1	1.167e-05 watt
Tdf2	-3.988e-12 ns
Tdr2	1.611e-11 ns
Average Power2	4.505e-07 watt
Peak Power2	1.251e-05 watt
Tdf3	-3.988e-12 ns
Tdr3	1.611e-11 ns
Average Power3	4.505e-07 watt
Peak Power3	1.251e-05 watt



ii. $B=1$ and A changes from 0 to 1.



t_df1	t_dr1	t_df2	t_dr2
t_df3	t_dr3	avg_power1	peak_power1
avg_power2	peak_power2	avg_power3	peak_power3
temper	alter#		
-1.924e-11	6.525e-11	-3.988e-12	1.611e-11
-3.988e-12	1.611e-11	1.362e-06	1.167e-05
4.505e-07	1.251e-05	4.505e-07	1.251e-05
25.0000	1		

Tdf1	-1.924e-11 ns
Tdr1	6.525e-11 ns
Average Power1	1.361e-06 watt
Peak Power1	1.167e-05 watt
Tdf2	-3.988e-12 ns
Tdr2	1.611e-11 ns
Average Power2	4.505e-07 watt
Peak Power2	1.251e-05 watt
Tdf3	-3.988e-12 ns
Tdr3	1.611e-11 ns
Average Power3	4.505e-07 watt
Peak Power3	1.251e-05 watt

